

What Remains of the Split-Forest Idea?

A Companion Note to the Generated Split-Forest Decidability Proof for $\mathcal{ALCI}_{\text{RCC5}}$

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Abstract

This note explains which parts of the original split-tree/split-forest proof strategy survive in the repaired decidability proof for $\mathcal{ALCI}_{\text{RCC5}}$. The short answer is: most of the architecture survives, but the objects must be interpreted correctly. The tree is a generated witness-cover presentation, not the Hasse diagram of an arbitrary semantic PP/PPI order. The finite object is a regular certificate, not necessarily a finite semantic model. Blocking repetition is not semantic equality. Sibling tables become support-closed mosaics. Under these clarifications, the containment backbone, split copies, sibling interfaces, the ordinary DR/PO frontier simplification, typed EQ quotienting, and finite certificate search all remain central parts of the proof.

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1. Executive summary

The repaired proof keeps the original split-forest idea, but gives it the right model-theoretic interpretation. The original idea should not be formulated as:

every satisfying RCC5 model has a finite rooted Hasse cover tree.

That formulation is too strong and unnecessary. The corrected formulation is:

every satisfiable concept has a finite regular generated split-forest certificate.

The unfolding of the certificate may be infinite. It may contain infinite PP- or PPI-rays. It need not have a semantic ambient top or bottom. Nevertheless, it is still tree-like: the relevant containment structure is represented by a generated tree of split occurrences.

The single most important reinterpretation is:

cover edge = selected generated witness-cover edge.

It is not:

cover edge = semantic Hasse cover in an arbitrary PP/PPI order.

Once this distinction is made, the earlier density/Hasse-cover worry disappears from the intended abstract Kripke/tableau setting. The proof constructs a sparse generated presentation using selected witnesses. It does not try to read an arbitrary semantic partial order as a Hasse diagram.

Original proof idea	Status in the repaired proof
Tree-shaped containment backbone	Survives as a generated witness-cover tree, not as a semantic Hasse diagram.
Split copies	Survive; they are needed for elements or obligations appearing in several containment contexts.
Sibling interfaces	Survive; they must be support-closed mosaics, not raw pair tables.
DR/PO sibling simplification	Survives for ordinary non-core sibling frontier nodes after containment, core, and equality constraints have been accounted for.
Typed EQ quotient	Survives; it must be an explicit congruence and must be kept separate from blocking.
Finite quotient	Survives as a finite regular certificate, not as a finite semantic model.

Thus the repaired proof is still recognizably a split-forest proof. What changes is the meaning of the words “cover”, “root”, “quotient”, and “table”.

2. Three levels of structure

The repaired proof separates three levels that were easy to conflate.

2.1. Semantic RCC5 structure

A semantic abstract RCC5 frame has a domain Δ and a total atomic relation map

$$r : \Delta^2 \rightarrow \{\text{EQ}, \text{PP}, \text{PPI}, \text{PO}, \text{DR}\}.$$

For strong equality semantics,

$$r(x, y) = \text{EQ} \iff x = y.$$

The labels must satisfy converse and composition constraints. In particular, PP and PPI are converse strict containment directions.

This semantic structure is not required to supply immediate covers. The proof does not depend on a geometric space, a topological space, or a semantic Hasse diagram.

2.2. Generated witness-cover structure

The proof first reduces to a witness-generated abstract model. For every active existential formula

$$\exists R.D \in \text{Cl}(C_0),$$

if a generated node a satisfies it, the construction chooses a witness b with

$$aRb \quad \text{and} \quad b \models D.$$

This choice gives a generated edge

$$a \xrightarrow{R}_{\text{Gen}} b.$$

That edge is immediate in the generated presentation, not necessarily in the semantic RCC5 relation.

For the containment-oriented split forest, selected PP/PPI witnesses are arranged vertically. With the convention that a parent is a superpart and a child is a proper part,

$$y \text{ is a parent of } x \implies x\text{PP}y.$$

The semantic PP/PPI relation along a branch is recovered by strict ancestor/descendant closure.

2.3. Finite regular certificates

The decision procedure searches over finite certificates. A finite certificate may contain cycles, but those cycles are blocking cycles. They denote infinitely many fresh occurrences in the unfolding.

For example,

$$u_0, u_1, \dots, u_m, (v_0, \dots, v_{k-1})^\omega$$

is a finite description of an infinite regular trunk. The repeated finite profiles are not identified by EQ. They are distinct semantic occurrences unless the explicit equality congruence says otherwise.

This distinction is crucial:

$$\text{blocking repetition} \neq \text{semantic EQ}.$$

3. The original split-forest thesis

The original proof idea had three conceptual components.

First, use a tree-shaped containment backbone. Along a branch, PP and PPI are represented by ancestor/descendant comparability. With the convention above, if y is a strict ancestor of x , then

$$xPPy,$$

and if y is a strict descendant of x , then

$$xPPIy.$$

Second, split occurrences when the semantic interaction is not tree-shaped. Splitting lets one give different presentation parents or different local contexts to occurrences that will later be synchronized by equality data when appropriate.

Third, store the remaining cross-subtree information in finite sibling interfaces. These interfaces describe how incomparable subtrees may be related by DR, PO, EQ, PP, or PPI, subject to the RCC5 composition table and the concept types at the endpoints.

All three components remain present in the repaired proof.

4. What survives essentially intact

4.1. The containment backbone

The backbone survives. The repaired proof still represents containment vertically. The difference is only that vertical edges are generated witness-cover edges rather than semantic Hasse edges.

This is the right reading for a tableau-style proof. An existential demand such as

$$\exists PPI.D$$

selects a generated PPI witness. In the containment orientation, that witness is placed as a proper-part child. Conversely, an existential demand such as

$$\exists PP.D$$

requires a superpart witness and is represented upward on the containment trunk.

The transitive closure of the branch supplies the semantic containment relations. Thus vertical comparability still carries the PP/PPI roles.

4.2. Split copies

Split copies also survive. They are still needed because a tree occurrence has one vertical parent, while the semantic situation may require an element or an obligation to be visible in several containment contexts. Splitting separates these contexts in the presentation.

A split copy is an occurrence. It is not automatically a separate semantic element, and it is not automatically equal to another occurrence. The final strong-equality model is obtained only after checking and quotienting the typed EQ congruence.

This is a direct continuation of the original split-forest intuition: make the presentation tree-like, then repair equality and cross-links explicitly.

4.3. Sibling interfaces

Sibling interfaces remain one of the main technical devices. Once containment is represented vertically, the remaining non-local constraints are between incomparable subtrees. The certificate therefore stores finite interface information between sibling regions of the generated tree.

The repair is that the interfaces must be support-closed mosaics. A raw pair table can lose joint witness information. For example, a relation between endpoint profiles may be witnessed by one occurrence, while a child transition from one of those profiles is witnessed by another occurrence. Combining those facts can fabricate support that is not jointly realizable.

A support-closed mosaic stores finite joint patterns and their restrictions. It records enough information to ensure that pair labels, child support, equality synchronization, and composition constraints can be realized together.

4.4. The ordinary DR/PO sibling frontier simplification

The DR/PO simplification survives in its correct scope. It should not be stated as “all sibling edges are DR/PO”. Core copies, equality mates, and composition-induced containment edges may force other labels.

The correct statement is:

after core, containment, and equality constraints are accounted for,

ordinary non-core sibling frontier edges can be chosen from $\{\text{DR}, \text{PO}\}$.

This remains a useful reason for the split construction: it isolates forced containment and equality effects so that the remaining open sibling frontier is simpler.

4.5. Typed equality quotienting

Typed EQ quotienting survives. The repaired proof makes it explicit.

There is an equality relation $\equiv_{\text{EQ}}$ on split occurrences. It may be quotiented only if it is a typed congruence. This requires at least:

- identical closure type;
- identical relevant witness obligations;
- synchronized external RCC5 labels to all third occurrences;
- compatibility with side mosaics and vertical summaries;
- no identification along a strict PP/PPI path.

After quotienting by $\equiv_{\text{EQ}}$, the resulting model satisfies strong equality:

$$r([x], [y]) = \text{EQ} \iff [x] = [y].$$

This is the formal version of the original idea that weak split copies can be synchronized and then collapsed.

4.6. Finite quotient search

The finite quotient idea survives as finite certificate search. Decidability is obtained by enumerating finite objects, not by assuming that every satisfiable concept has a finite semantic model.

A valid certificate contains finite data for:

- local closure types;
- containment profiles;
- context transitions;
- finite reachability summaries for PP/PPI eventualities;
- support-closed sibling mosaics;
- typed equality congruence;
- cycle acceptance conditions for regular trunks.

The unfolding of such a certificate may be infinite, but it is regular and effectively described.

5. What was revised

5.1. Generated cover, not semantic Hasse cover

The most important revision is to replace semantic Hasse covers by generated witness covers. The proof now says:

$$E_{PP}(x, y) \implies xPPy,$$

where E_{PP} is a generated edge. It does not say that y is the immediate semantic PP-successor of x .

This removes the need to discuss dense semantic orders. Density is simply not relevant to the generated abstract presentation used by the decision proof.

5.2. Pointed regular trunks, not ambient semantic roots

The generated split forest is pointed at the evaluation occurrence, but it need not terminate at a semantic top. A formula such as

$$\exists PP.T \sqcap \forall PP.\exists PP.T$$

requires an infinite chain of superparts. In the containment orientation, this is an infinite upward trunk. The finite certificate represents it by a blocking cycle, whose unfolding gives fresh occurrences:

$$a_0PPa_1PPa_2PP\cdots.$$

There is no semantic root at the top of this chain.

5.3. Cycle acceptance, not explicit Buechi machinery

The repaired original-style proof does not need to introduce Buechi automata as a separate object. It does need the corresponding finite eventuality condition.

For instance, if a cycle node carries an obligation

$$\exists PP.D,$$

then some later node on the repeated cycle must satisfy D . The target may be the same profile on the next lap, because the next lap is a fresh semantic occurrence. Similarly, universal PP/PPI requirements must be propagated along all strict later occurrences in the relevant direction.

Thus the automata-theoretic idea is present only as a finite cycle acceptance check.

5.4. Mosaics, not raw relation tables

The original pair-table intuition survives only after strengthening. The certificate must not aggregate independent pair facts in a way that allows false amalgamation. It must store support-closed joint mosaics.

A mosaic records a finite local pattern of profiles and pair labels, closed under restrictions and under the support operations needed by the RCC5 composition table. This is what lets the soundness proof assign one coherent label to every relevant pair in the unfolding.

5.5. Blocking and equality are separate

The original quotient terminology can be misleading. Blocking identifies finite descriptions, not semantic elements. Equality quotienting identifies semantic elements, but only through the explicit typed EQ congruence.

So the repaired proof uses two separate notions:

$\equiv_{\text{Blk}}$ for repeated finite profiles,

$\equiv_{\text{EQ}}$ for semantic equality.

Only $\equiv_{\text{EQ}}$ is used to quotient the semantic unfolding.

6. What does not survive

The following stronger claims should not be retained.

1. **Semantic Hasse diagram.** The split forest is not extracted as the Hasse diagram of an arbitrary semantic PP/PPI relation.
2. **Ambient root.** The containment presentation need not have a semantic top or bottom element.
3. **Well-foundedness of semantic containment.** The proof uses generated witness closure and acyclicity, not a global well-foundedness assumption.
4. **Raw pair-table completeness.** Pairwise labels indexed only by endpoint type are not enough for completeness and soundness.
5. **Repeated state equals EQ.** A blocking cycle unfolds to fresh occurrences; it is not an equality cycle.
6. **Finite semantic model property.** The finite object is a certificate. The model it denotes may be infinite.

These are not losses of the core idea. They are removals of assumptions that were stronger than the proof needs.

7. How the coherent proof now proceeds

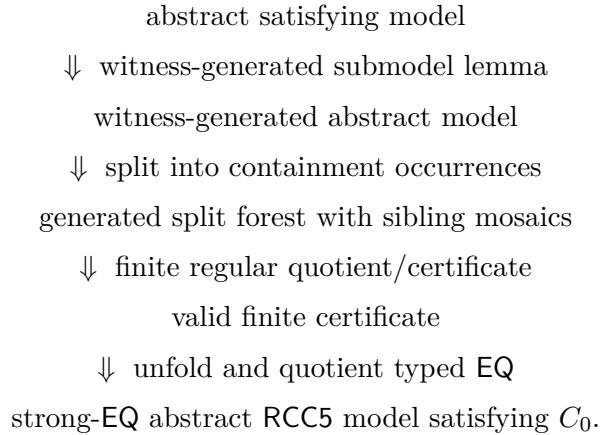
The repaired proof can still be organized in the original split-forest style.

1. Build the closure $\text{Cl}(C_0)$ with NNF complements and define Hintikka types.
2. Prove the witness-generated submodel lemma: any satisfying abstract model can be thinned to one generated by selected existential witnesses from $\text{Cl}(C_0)$.
3. Arrange selected PP/PPI witnesses into a containment-oriented generated split forest.
4. Use split copies to keep the containment presentation tree-like.
5. Define finite profiles recording local type, vertical summaries, side-context data, equality colour, and status.
6. Define support-closed sibling mosaics for cross-subtree labels and DR/PO witnesses.
7. Define finite validity checks: local type consistency, vertical propagation, eventuality realization, mosaic support, RCC5 composition, and typed EQ congruence.
8. Prove soundness by unfolding a valid finite certificate and quotienting by the typed equality congruence.
9. Prove completeness by extracting such a finite certificate from a witness-generated split presentation.
10. Enumerate finite certificates to decide satisfiability.

This is a direct split-forest proof, not merely a list of patches.

8. The final picture

The repaired proof has the following conceptual pipeline:



The original insight is the middle part: containment is made vertical, and non-vertical interaction is stored in finite interfaces. The repaired proof adds the generated-model reduction before it and the precise regular-certificate semantics after it.

9. Conclusion

With the density/Hasse issue removed by the witness-generated submodel lemma, more of the original proof architecture remains intact than an initial semantic-Hasse reading would suggest. The surviving result is best stated as:

$\mathcal{ALCI}_{\text{RCC5}}$ has a generated regular split-forest model property.

This is weaker than a finite rooted semantic Hasse-cover model property, but it is exactly the kind of structure needed for a decidability proof. The original split-forest idea survives as the central organizing principle: vertical generated containment plus finite support-closed sibling interfaces, followed by typed equality quotienting and finite regular certificate search.